

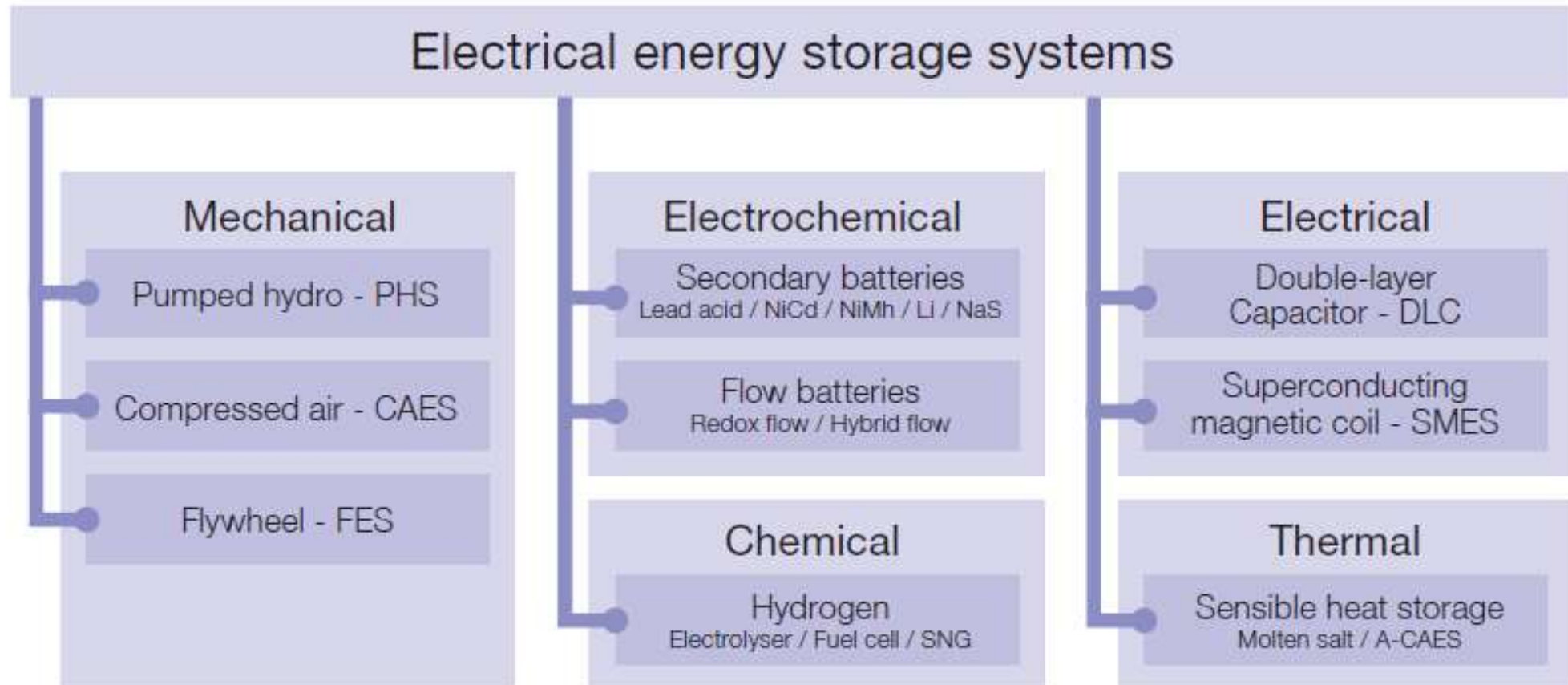
UNIT- III

FEATURES OF ENERGY STORAGE SYSTEMS



CLASSIFICATION OF EES SYSTEMS

- A widely-used approach for classifying EES systems is the determination according to the form of energy used.



MECHANICAL STORAGE SYSTEMS

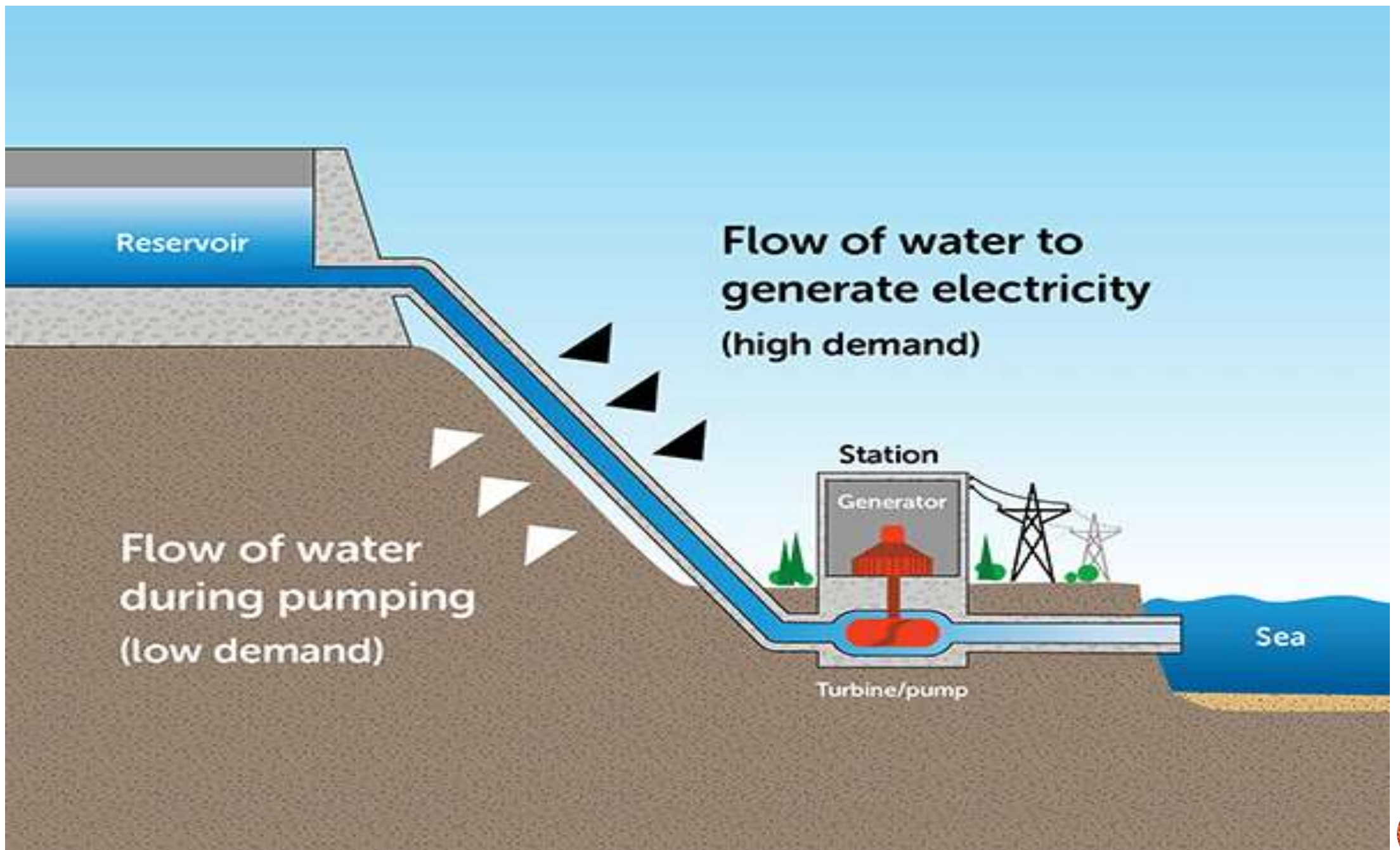
1. Pumped Hydro Storage (PHS)
2. Compressed Air Energy storage (CAES)
3. Flywheel Energy Storage (FES)



PUMPED HYDRO STORAGE (PHS)

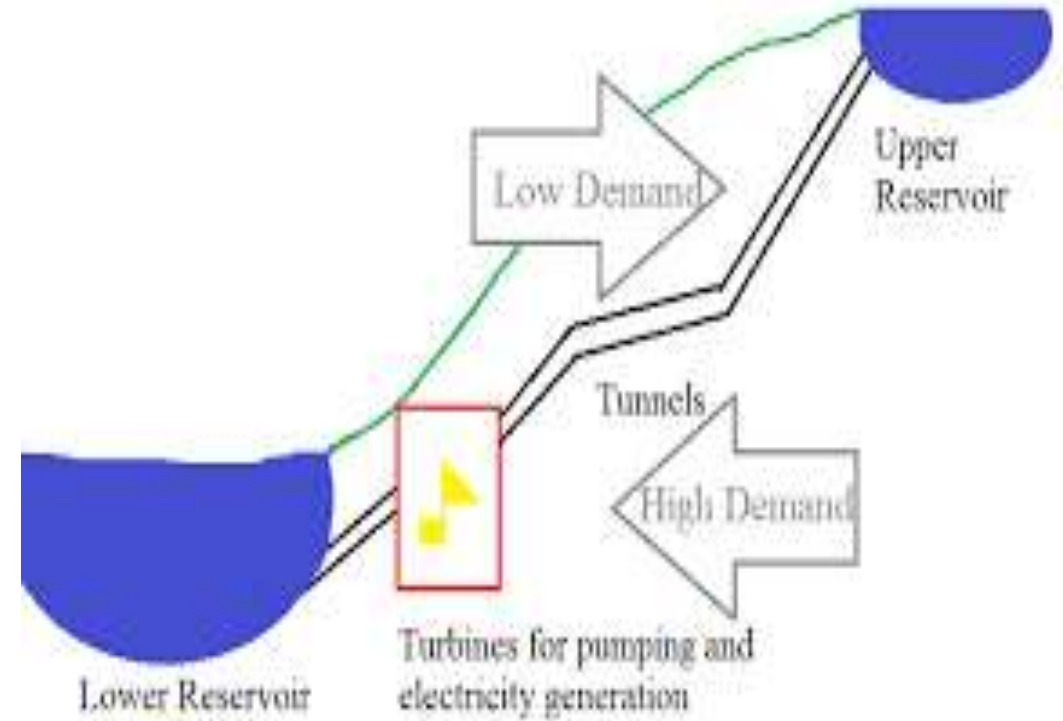
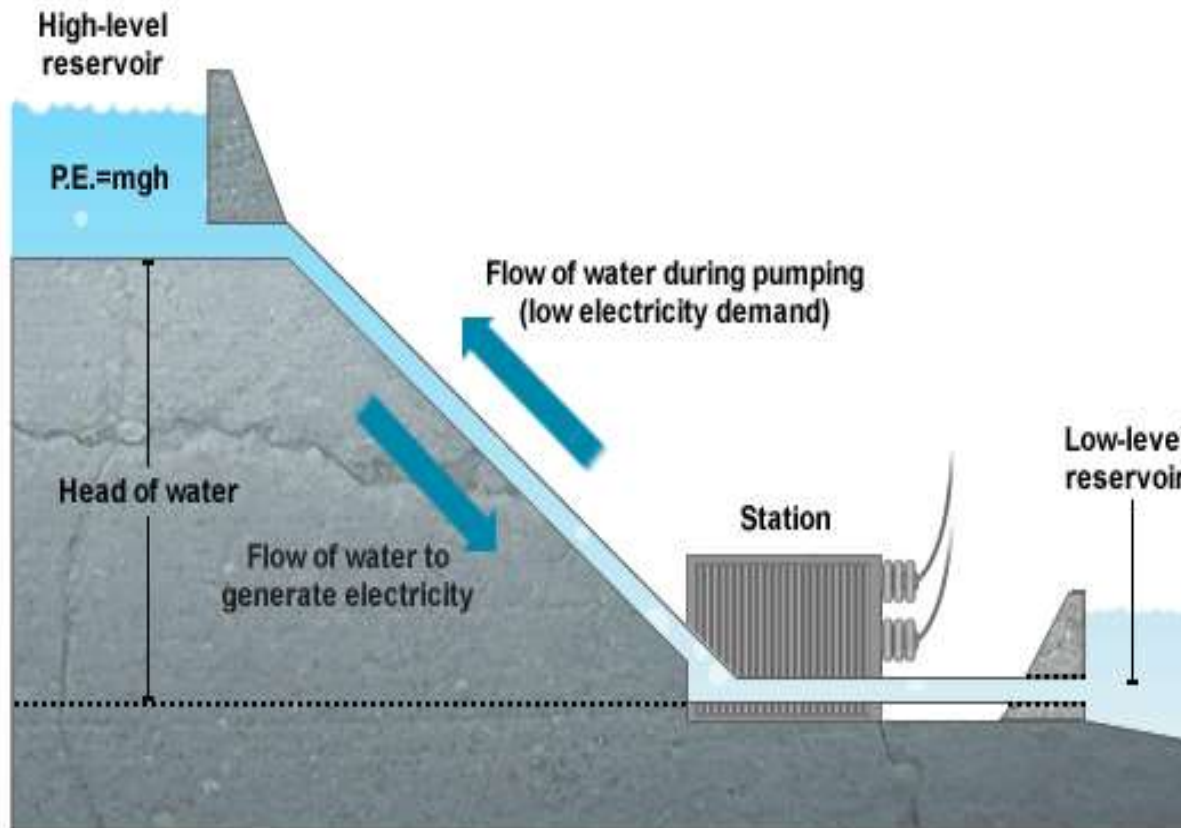
- World wide – 120GW (3% of global generation capacity)
- First PHS plant - Italy and Switzerland in the 1890's
- Largest PHS plant in the world – Bath country pumped storage station with 3003MW capacity located in Virginia, USA
- In India – Sardar Sarovar Dam, 1200MW on Narmada river near Navagam, Gujarat





- This method stores energy in the form of gravitational potential energy of water, pumped from a lower elevation reservoir to a higher elevation.
- PHS uses two water reservoirs at different elevations. High dams can be used as PHS plants, for the lower reservoir flooded mine shafts, other underground cavities and the open sea will be used.
- Pumping from lower to upper – charging(During off peak hours)
- Flow back from upper to lower reservoir – discharging





- Reversible turbine-generator assemblies act as a combined pump & turbine generator unit > usually a **Francis turbine**
- Revenue increases by selling more electricity during periods of peak demand. Discharge time range from several hours to few days.
- Long service life up to - 75 years or more.
- Efficiency – 70% to 85%
- Capital costs relatively high and large land use.





PHS



COMPRESSED AIR ENERGY STORAGE (CAES)

- 440 MW of installations exist around the world.
- This technology is known & used since 19th century for different industrial applications including mobile ones.
- The 290MW CAES plant built in 1978 located in Huntorf, Germany.
- Air is used as storage medium due to its availability.

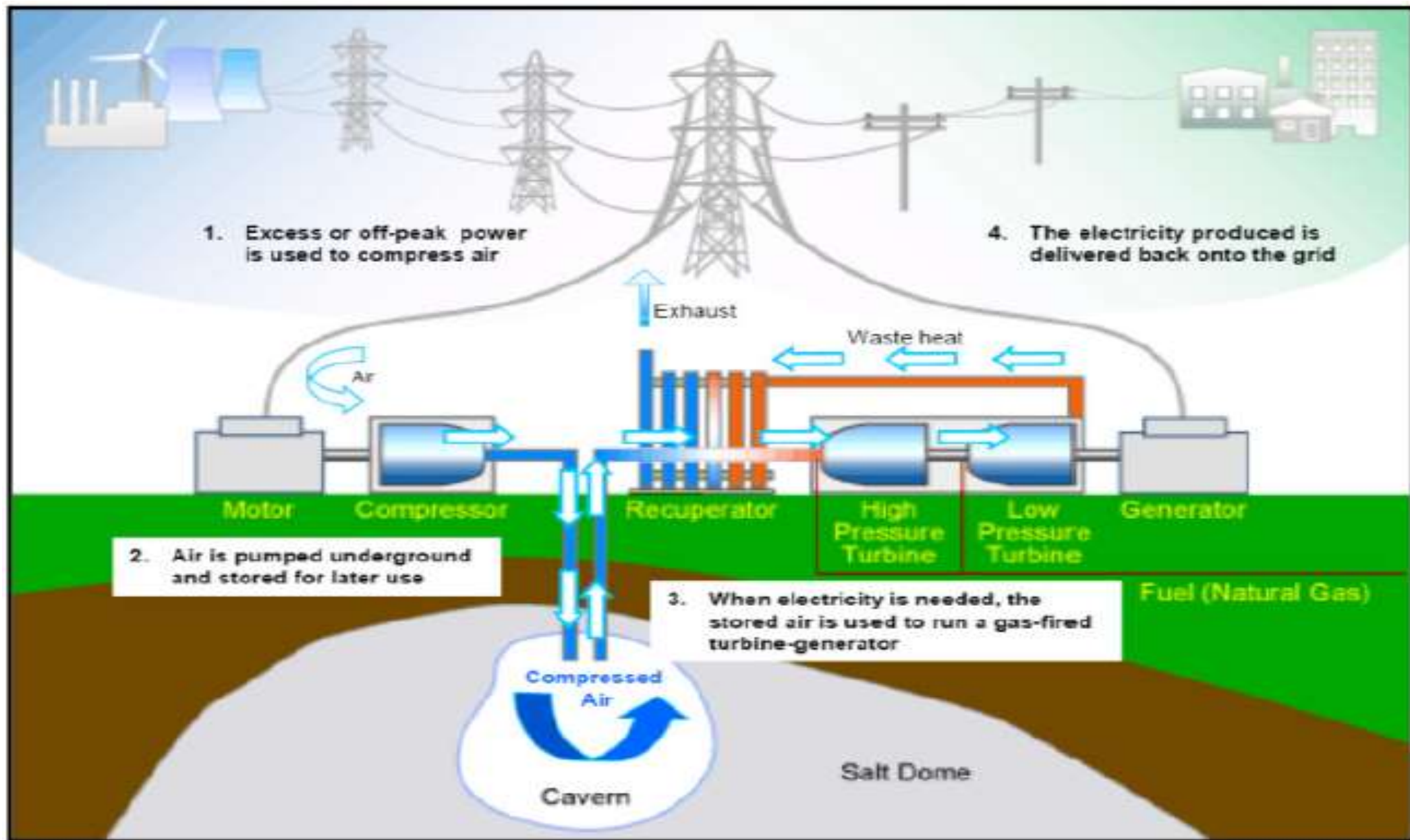


- This technology is based on conventional gas turbines.
- At off peak demands electricity is used to compress air and stores it in either an underground structure or on above-ground systems  **Charging**
- When needed the compressed air is mixed with natural gas, burned & expanded in a modified gas turbine  **discharging**



- Underground storage options are caverns, aquifers or abandoned mines.
- Large Capacity
- Low efficiency-less than 50% and geographic limitation of locations.





FLYWHEEL ENERGY STORAGE (FES)

- In flywheel energy storage rotational energy is stored in an accelerated rotor, a massive rotating cylinder.
- Flywheels were first used in steam engines.
- In the 1970s, when the technology was used for centrifugal enrichment of uranium, it was rediscovered as a potential new energy storage system.



- They store energy in a spinning mass.
- Through a coupled electric machine, the system can store power by accelerating the shaft, and retrieve power by slowing it.
- The energy will depend on the moment of inertia and the square of the angular speed of the flywheel shaft

$$E = \frac{1}{2} I \omega^2$$



- The energy storage capability of flywheels can be improved by using the right material and increasing the moment of inertia, or spinning it at higher rotational velocities.
- Flywheel energy storage systems have capacities from 0.5 to 500 kWh

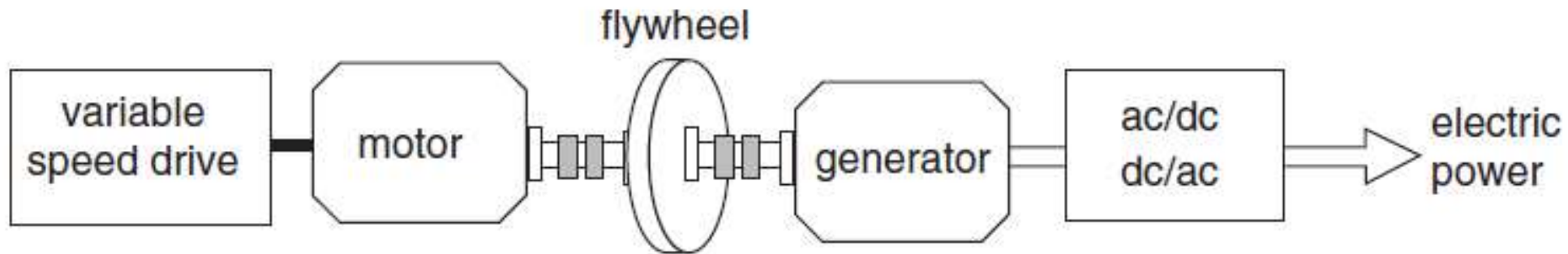
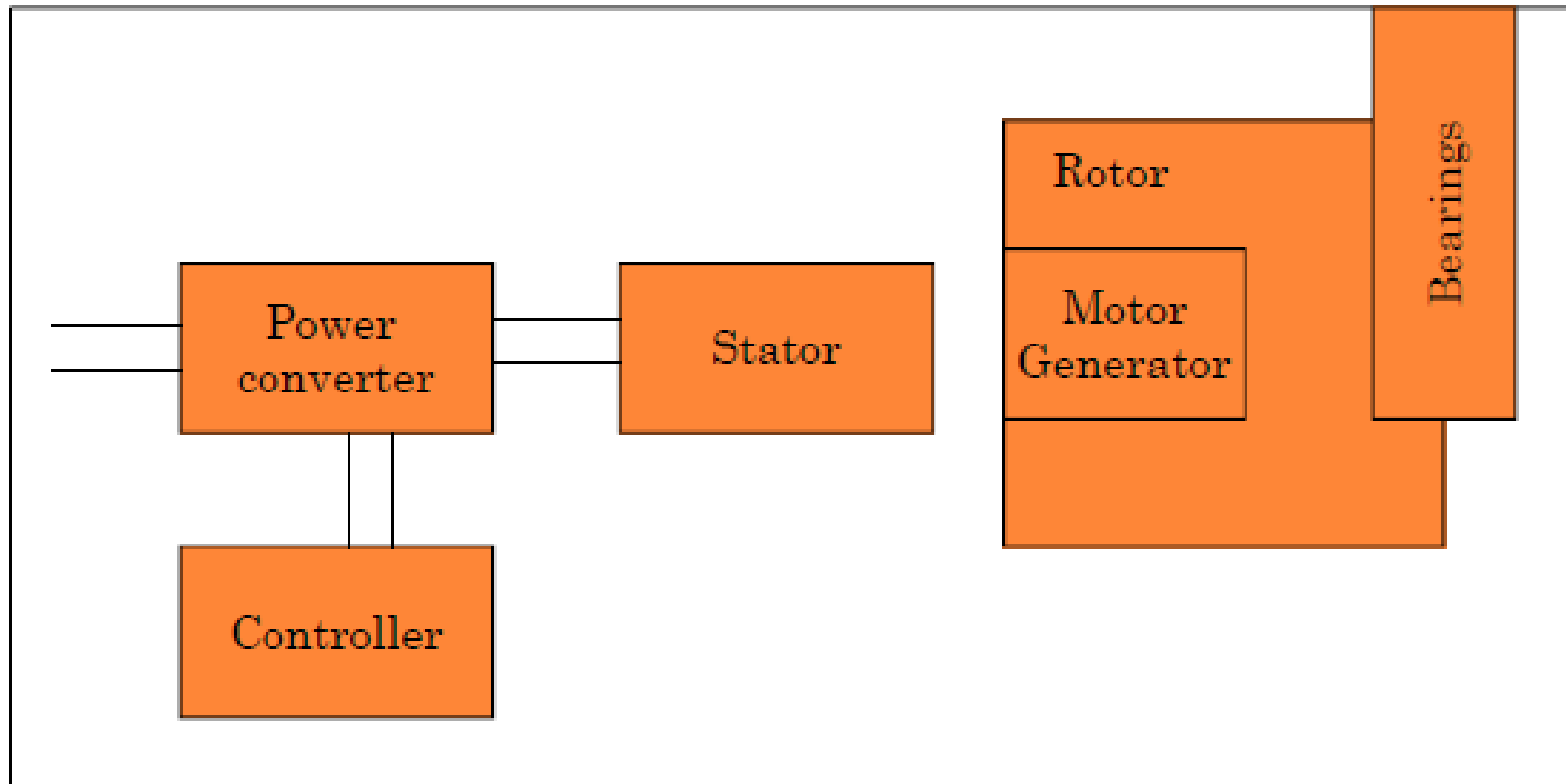


Fig. Standard flywheel system with motoring and generating sets

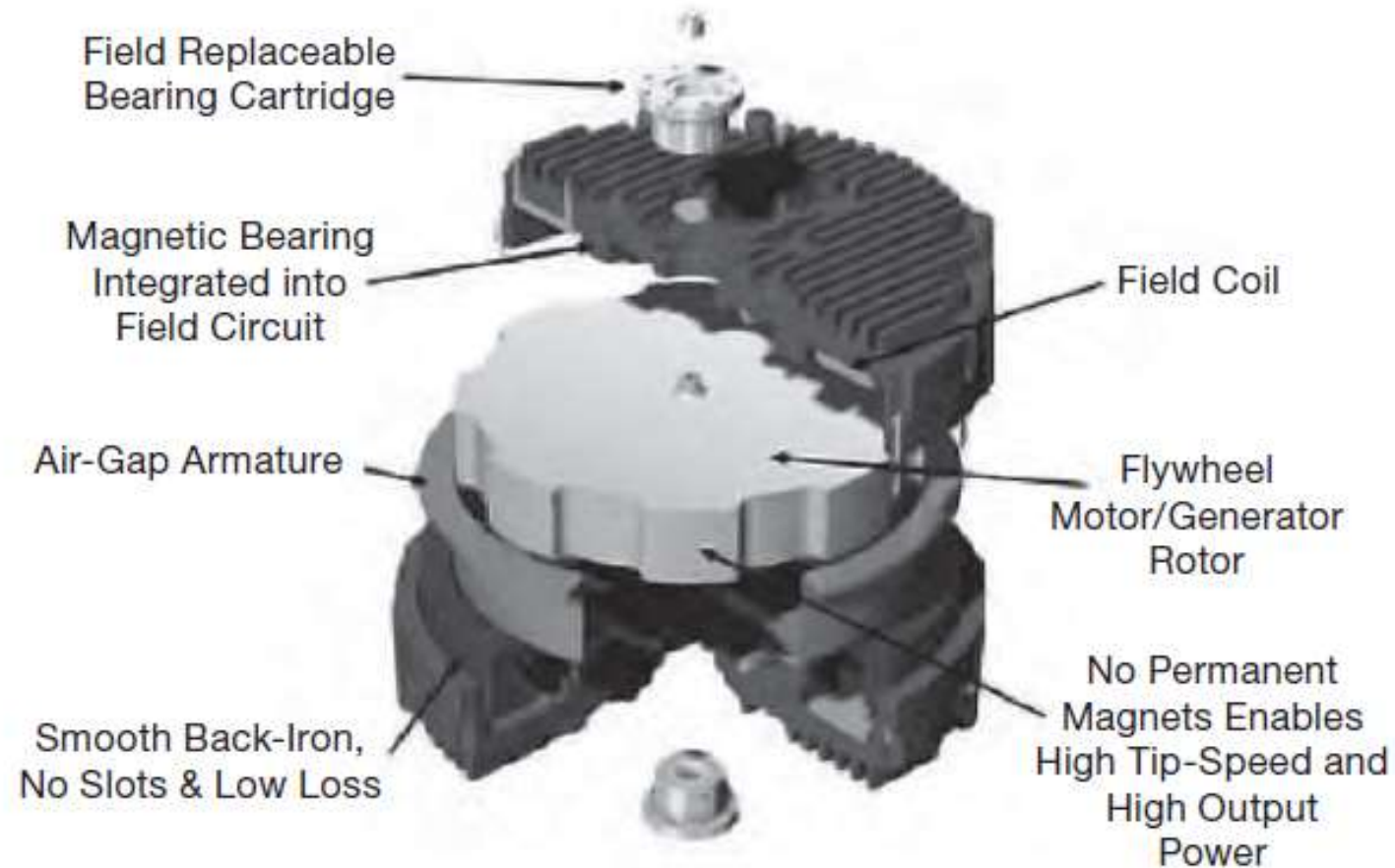




Schematic diagram of Flywheel subsystems



- The main components of flywheel are
 - The rotating body/cylinder (comprised of rim and shaft) in a compartment.
 - The bearings and the transmission device (motor/generator mounted onto the stator).



- The energy is maintained in the flywheel by keeping the rotating body at a constant speed.
- An increase in the speed results in a higher amount of energy stored.
- To accelerate the flywheel electricity is supplied by a transmission device.
- If the flywheel's rotational speed is reduced electricity may be extracted from the system by the same transmission device.



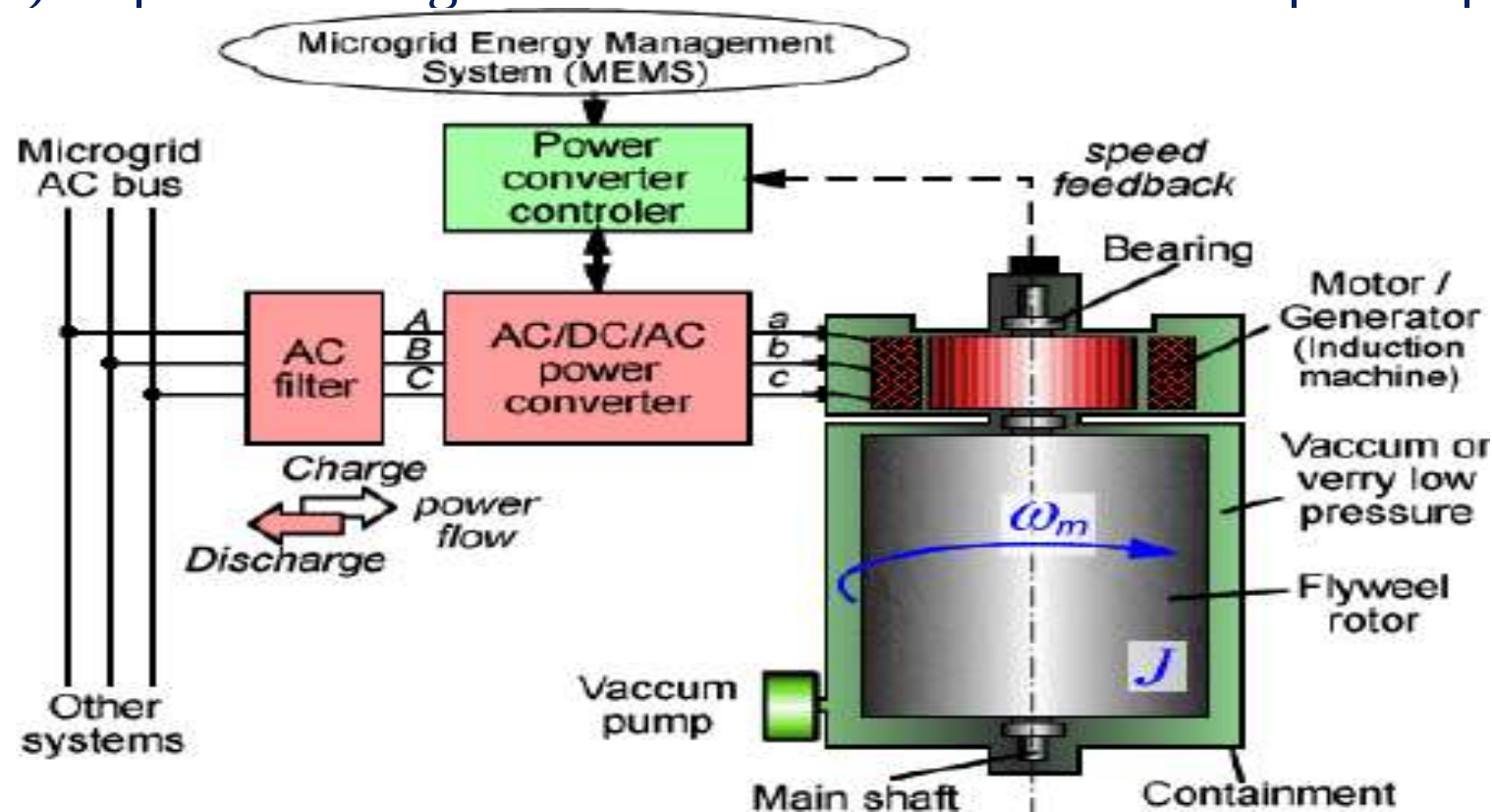
- First generations flywheels available since 1970's, use a large steel rotating body on mechanical bearings.
- Advanced FES systems have rotor made of high-strength carbon filaments, suspended by magnetic bearings, and spinning at speeds from 20000 to 50000 rpm in a vacuum enclosure.



- The main features of flywheel are
 - Excellent cycle stability
 - Long life
 - Little maintenance
 - High power density and
 - Use of environmentally inert material
- However, flywheels have a high level of self-discharge due to air resistance and bearing losses and suffer from low current density.



- Today flywheels are commercially deployed for power quality in industrial and UPS applications, mainly in a hybrid configuration.
- Efforts are being made to optimize flywheels for long-duration operation (up to several hours) as power storage devices for use in vehicles and power plants.



ELECTRO CHEMICAL STORAGE SYSTEMS

- (1) Secondary Batteries
- (2) Flow Batteries



1)SECONDARY BATTERIES

- Secondary batteries are six types :

(i)Lead Acid

(ii)NiCd / NiMH,

(iii)Li-ion

(iv)Metal Air

(v)Sodium Sulphur

(vi)Sodium Nickel Chloride



■ (i)Lead Acid battery (LA)

- Lead acid batteries are the world's most widely used battery type and have been commercially deployed since about 1890.
- Lead acid battery systems are used in both mobile and stationary applications. Their typical applications are emergency power supply systems, stand-alone systems with PV, battery systems for mitigation of output fluctuations from wind power and as starter batteries in vehicles.
- Disadvantages of lead acid batteries are usable capacity decrease when high power is discharged, lower energy density and the use of lead, a hazardous material prohibited or restricted in various jurisdictions.
- Advantages are low cost/performance ratio, easy recyclability and a simple charging technology.



■ (ii) Nickel Cadmium and Nickel Metal Hydride battery (NiCd, NiMH)

- Nickel cadmium (NiCd) batteries had been in commercial use since about 1915 and nickel metal hydride (NiMH) batteries around 1995.
- Compared to lead acid batteries, nickel-based batteries have a higher power density, a slightly greater energy density and the number of cycles is higher.
- NiCd batteries are the only batteries capable of performing well even at low temperatures in the range from -20 °C to -40 °C. Since 2006 they have been prohibited for consumer use, because of the toxicity of cadmium.
- NiMH batteries were developed initially to replace NiCd batteries. These batteries have much higher energy densities, In portable and mobile applications sealed NiMH batteries have been extensively replaced by lithium ion batteries, hybrid vehicles available on today's market operate almost exclusively with sealed NiMH batteries, as these are robust and far safer than lithium ion batteries, NiMH batteries currently cost about the same as lithium ion batteries.
- Maximal rated capacity which is ten times less when compared to NiCd and lead acid batteries.



■ (iii)Lithium Ion (Li-ion) battery

- Lithium ion batteries have become the most important storage technology in the areas of portable and mobile applications (e.g. laptop, cell phone, electric bicycle, electric car) since around 2000.
- High cell voltage levels of up to 3.7 nominal volts, one lithium ion cell can replace three NiCd or NiMH cells which have a cell voltage of only 1.2 Volts. advantage of Li-ion batteries is their high energy density(Weight/KG) and the prospect of large cost reductions through mass production.
- Lithium ion batteries have a very high efficiency, typically in the range of 95 % - 98 %.
- Safety is a serious issue in lithium ion battery technology. Most of the metal oxide electrodes are thermally unstable and high temperatures can lead to a thermal runaway.
- To minimize this risk, lithium ion batteries are equipped with a monitoring unit to avoid over-charging and over- discharging. Lithium ion battery technology is still developing, and there is considerable potential for further progress.



■ (iv) Metal air battery (Me-air)

- A metal air electrochemical cell consists of the anode made from pure metal and the cathode connected to an inexhaustible supply of air.
- Among the various metal air battery chemical couples, the lithium air battery is most attractive since its specific energy is 100 times more than other battery types. However, the high reactivity of **lithium with air and humidity can cause fire, which is a high safety risk.**
- Currently only a zinc air battery with a theoretical specific energy is technically feasible.
- Rechargeable zinc air cells have a **difficulty in design** since zinc precipitation from the water-based electrolyte must be closely controlled.
- A satisfactory, electrically rechargeable metal air system potentially offers low materials cost and high specific energy, but **none has reached marketability yet.**



■ (v) Sodium sulphur battery (NaS)

- Sodium sulphur batteries consist of liquid (molten) sulphur at the positive electrode and liquid (molten) sodium at the negative electrode; the active materials are separated by a solid beta alumina ceramic electrolyte. **The battery temperature is kept between 300 °C and 350 °C to keep the electrodes molten.**
- These batteries are efficient, efficiency is about 75 % and have fast response. NaS batteries to be economically used in combined power quality and time shift applications with high energy density.
- The main drawback is that **to maintain operating temperatures a heat source is required**, which uses the battery's own stored energy, partially reducing the battery performance.
- As the response time is in the range of milliseconds and NaS batteries meet the requirements for grid stabilization, this technology will be very useful for utilities and large consumers.



■ (vi) Sodium Nickel chloride battery (NaNiCl)

- The sodium nickel chloride (NaNiCl) battery, better known as the **ZEBRA (Zero Emission Battery Research Activities) battery**, is – like the NaS battery – a high-temperature (HT) battery, and has been commercially available since about 1995. Its operating temperature is around 270 °C, and it uses nickel chloride instead of sulphur for the positive electrode.
- NaNiCl batteries can withstand limited overcharge and discharge and have potentially better safety characteristics and a higher cell voltage than NaS batteries.
- These batteries have been successfully implemented in several **electric vehicle designs**.
- Present research is in developing advanced versions of the ZEBRA battery with **higher power densities for hybrid electric vehicles**, and also high-energy versions for storing renewable energy for load-levelling and industrial applications.



Table 3.1 Comparison of Different Battery Energy Storage Systems

	Lead acid	Nickel cadmium	Sodium sulphur	Lithium ion	Sodium nickel chloride
Achieved/demonstrated upper limit power	Multiple tens of MW	Tens of MW	MW scale	Tens of kW	Tens/low hundreds of kW
Specific energy (Wh/kg)	35 to 50	75	150 to 240	150 to 200	125
Specific power (W/kg)	75 to 300	150 to 300	90 to 230	200 to 315	130 to 160
Cycle life (cycles)	500 to 1500	2,500	2,500	1,000 to 10,000+	2,500+
Charge/discharge energy efficiency (%)	~80	~70	up to 90	~95	~90
Self discharge	2 to 5% per month	5 to 20% per month	#	~1% per month	#



2)FLOW BATTERIES



- In conventional secondary batteries, the energy is charged and discharged in the active masses of the electrodes.
- A flow battery is also a rechargeable battery, but the energy is stored in one or more electro active species which are dissolved in liquid electrolytes.
- The electrolytes are stored externally in tanks and pumped through the electrochemical cell that converts chemical energy directly to electricity and vice versa.



2)FLOW BATTERIES

- **(i) Redox Flow Battery (RFB)**
- **(ii)Hybrid Flow Battery (HFB)**

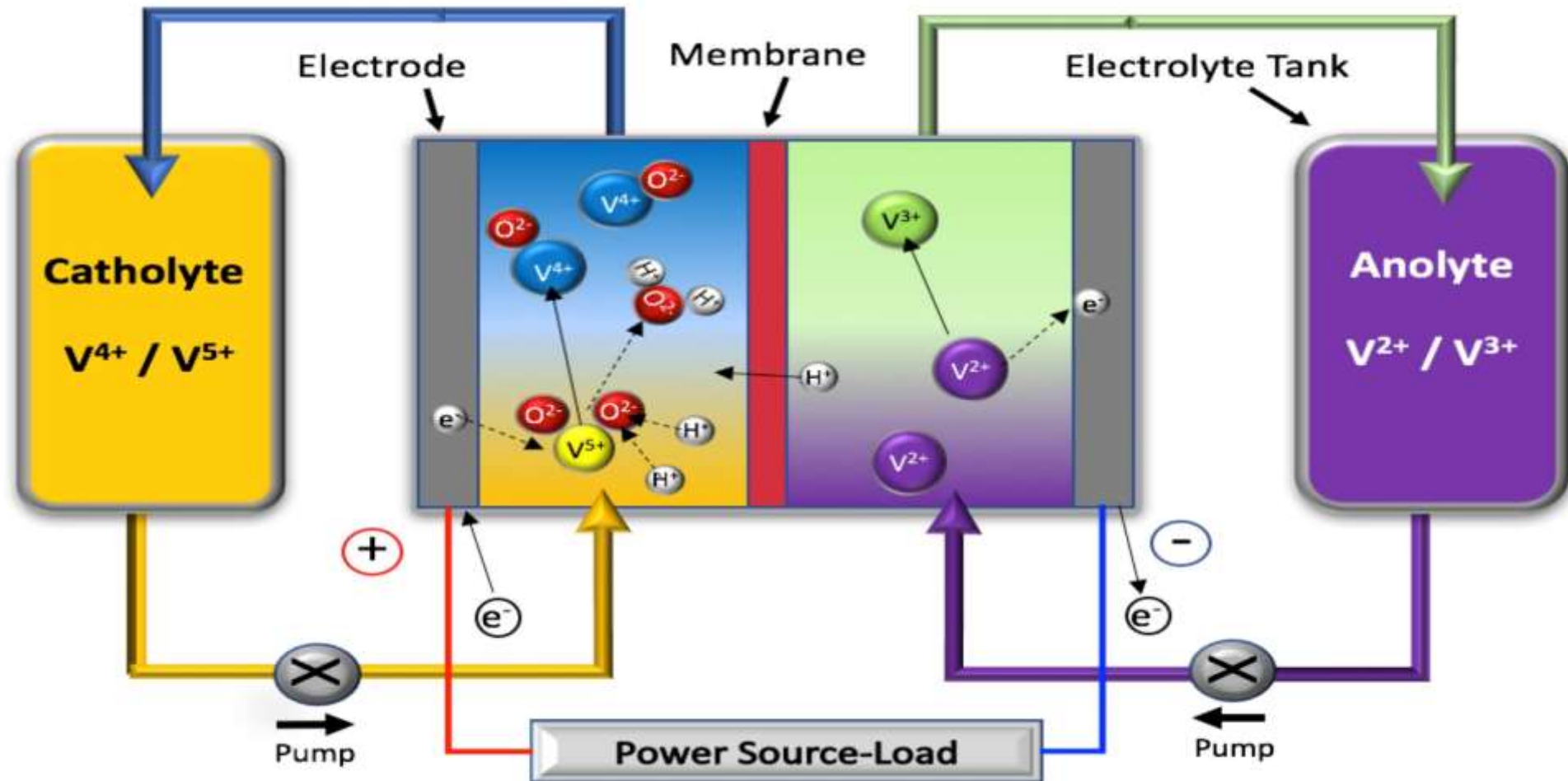


(I) REDOX FLOW BATTERY (RFB)

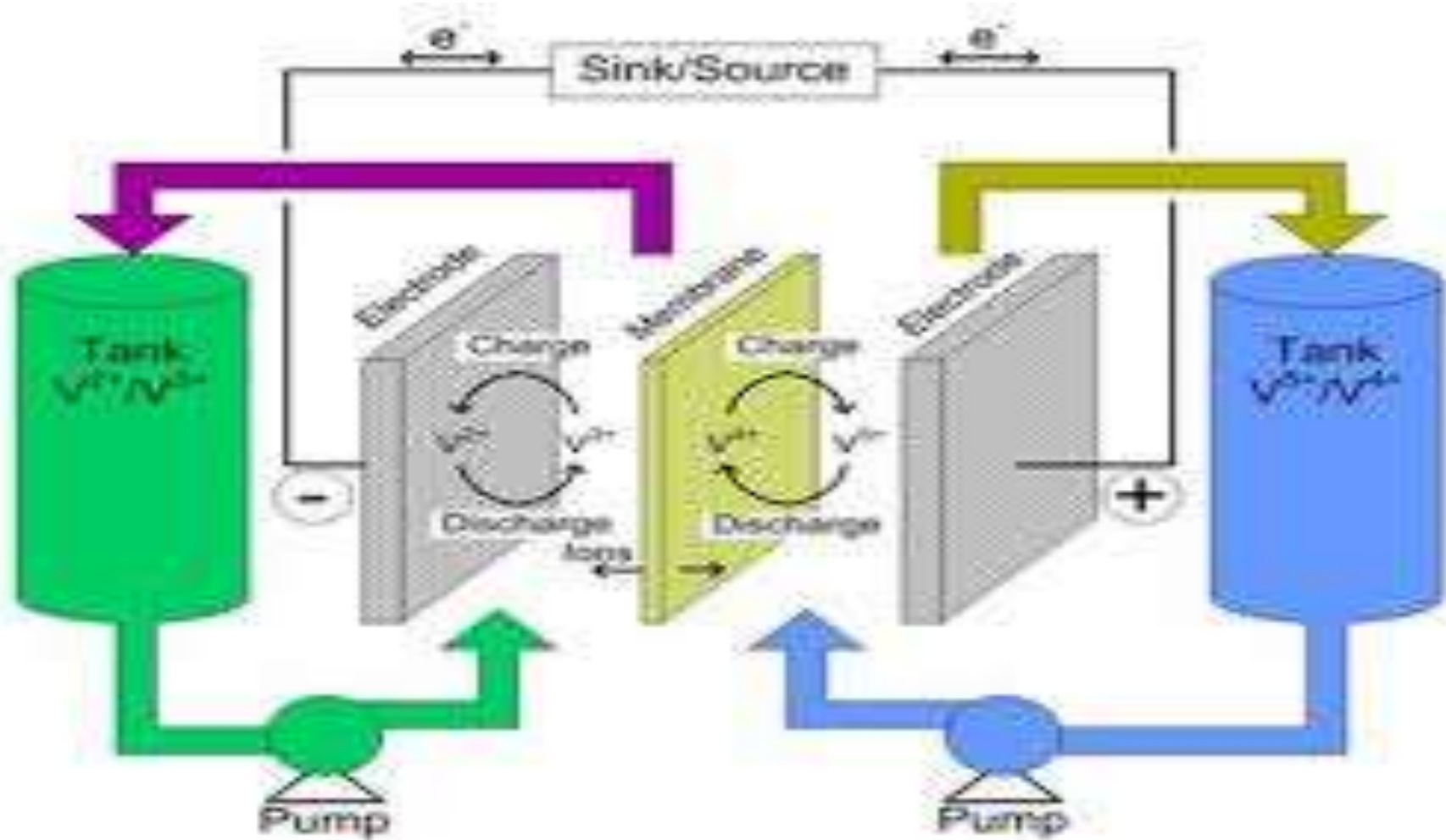
- In redox flow batteries (RFB) two liquid electrolyte dissolutions containing dissolved metal ions as active masses are pumped to the opposite sides of the electrochemical cell.
- The electrolytes at the negative and positive electrodes are called anolyte and catholyte respectively. Today various redox couples have been investigated and tested in RFBs, such as a Fe- Ti system, a Fe-Cr system and a poly S-Br system. The Vanadium redox flow battery has been developed.
- During charging and discharging the metal ions stay dissolved in the fluid electrolyte as liquid.
- The main advantage of this battery is the use of ions of the same metal on both sides.



SCHEMATIC OF A VANADIUM REDOX FLOW BATTERY



VANADIUM REDOX FLOW BATTERY



- The VRFB uses a V^{2+}/V^{3+} redox couple as oxidizing agent and V^{5+}/V^{4+} redox couple as reducing agent in mild sulphuric acid.
- The VRFB storage system was pioneered at the University of New South Wales, Australia in the early 1980s.
- A VRFB storage system of up to 500kW and 10hrs has been installed in Japan by SEI. It also used a VRFB in power quality applications.



II)HYBRID FLOW BATTERY (HFB)

- In a hybrid flow battery (HFB) one of the active masses is internally stored within the electrochemical cell, whereas the other remains in the liquid electrolyte and is stored externally in a tank. Therefore hybrid flow cells combine features of conventional secondary batteries and redox flow batteries.
- Typical examples of a HFB are the Zn-Ce and the Zn-Br systems. In both cases the anolyte consists of an acid solution of Zn^{2+} ions. During charging Zn is deposited at the electrode and at discharging Zn^{2+} goes back into solution.



CHEMICAL ENERGY STORAGE

- Chemical energy storage focuses on hydrogen and synthetic natural gas (SNG) as secondary energy carriers, since these could have a significant impact on the storage of electrical energy in large quantities.
- The main purpose of such a chemical energy storage system is to use “excess” electricity to produce hydrogen via water electrolysis.
- Chemical energy storage is the only concept which allows storage of large amounts of energy, up to the TWh range, and for greater periods of time – even as seasonal storage.
- Another advantage of hydrogen and SNG is that these universal energy carriers can be used in different sectors, such as transport, mobility, heating and the chemical industry.



Hydrogen (H₂)

- A typical hydrogen storage system consists of an electrolyzer, a hydrogen storage tank and a fuel cell.
- An electrolyzer is an electrochemical converter which splits water with the help of electricity into hydrogen and oxygen.
- Hydrogen is stored under pressure in gas bottles or tanks, and this can be done practically for an unlimited time.
- To generate electricity, both gases flow into the fuel cell where an electrochemical reaction which is the reverse of water splitting takes place: hydrogen and oxygen react and produce water, heat is released and electricity is generated.
- For economic and practical reasons oxygen is not stored but vented to the atmosphere on electrolysis, and oxygen from the air is taken for the power generation. The overall AC-AC efficiency is around 40 %.



Synthetic Natural Gas (SNG)

- Synthesis of methane (also called synthetic natural gas, SNG) is the second option to store electricity as chemical energy.
- The water splitting process in an electrolyzer gives hydrogen, this hydrogen and carbon dioxide reaction will give methane in a methanation reactor.
- The SNG produced can be stored in pressure tanks, underground, or fed directly into the gas grid. Several CO₂ sources are conceivable for the methanation process, such as fossil-fuelled power stations, industrial installations or biogas plants.
- The production of SNG is preferable at locations where CO₂ and excess electricity are both available. In particular, the use of CO₂ from biogas production processes is promising as it is a widely-used technology.



- Methane has a higher energy density, and transport in pipelines requires less energy (higher density of the gas).
- The main disadvantage of SNG is the relatively low efficiency due to the conversion losses in electrolysis, methanation, storage, transport and the subsequent power generation.
- The overall AC-AC efficiency, < 35 %, is even lower than with hydrogen.



Chemical Energy Storage

